EGM 6365 - Structural Optimization Project

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***Abstract*—The objective of the project was to find the optiThe 99-line topology optimization MATLAB code written by O. Sigmund was utilized to first create the initial conceptual design. The design was created in Ansys Workbench. Using Ansys Mechanical, Ansys's Finite Element Analysis (FEA) software, the forces, structural dimensions, and material properties were inputted and the simulation was run. A parametric design study was conducted with four dimensions of the structure being chosen as the design parameters. Latin Hypercube Sampling (LHS) was used to create samples of the design parameters, and the simulation was run with the new geometries. The sample outputs were recorded along with the input design parameters. These samples were imported into MATLAB for surrogate modeling. The Kriging and Polynomial Response Surface (PRS) surrogate methods used the FEA data to predict the response of the structure given the design parameters, and the output parameters of weight were minimized while the maximum experienced von Mises stress were maximized. An FEA element size of 0.025m was used for each simulation. The initial mass was found to be 281 kg with a maximum von Mises stress of 43.6 MPa. From Kriging simulations, the final mass was reduced to 184.9 kg, 34.2% lighter, and a maximum experienced von Mises stress 73.885 MPa, a 36.4% increase. From PRS simulations the final mass was reduced to 183.3 kg, 34.8% lighter, and a maximum experienced von Mises stress 72.314 MPa, a 35% increase.**

***Index Terms*— Ansys, finite element analysis (FEA), finite element method (FEM), topology optimization**

I. Introduction

Topology optimization is a method to determine an object's optimal design given a specific set of criteria and constraints that the design needs to achieve. This method uses algorithms to add and remove material within a specified design space. It uses the finite element analysis (FEA) model, and iteratively calculates each element to see if the element experiences loading. If not, then the algorithm removes the element and thus the material is slowly optimized by this repetition.

Messerschmitt-Bölkow-Blohm Beam

The half Messerschmitt-Bölkow-Blohm (MBB) beam is a beam model that is often used for topology optimization derived from a larger beam model. The half MBB beam model is a simplification of a larger beam where the beam is assumed to have constant material properties, and the force placement is symmetrical from the perspective of the midpoint of the original beam mode. The full beam is often represented with a concentrated force in the middle, as shown in Figure 1. The MBB beam is often represented as half of the full beam as a rectangle with the left vertical face and the right bottom tip of the beam being supported by rollers, with a concentrated force at the top left corner as shown in Figure 2.

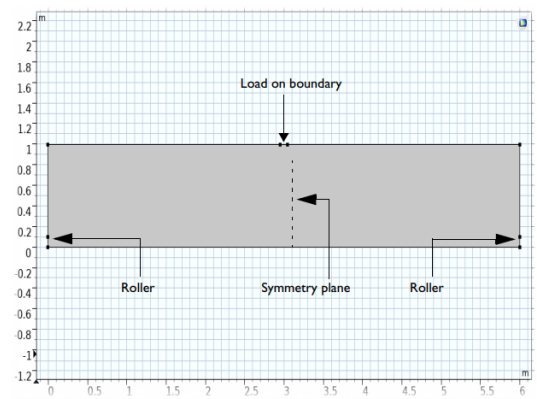

Fig. 1. Beam model with concentrated load at the midpoint of the beam with two roller supports at the very ends of the beam length *[1]*.

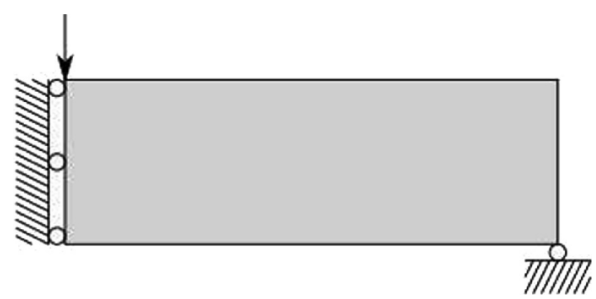

Fig. 2. Messerschmitt-Bölkow-Blohm (MBB) Beam *[2]*.

Latin Hypercube Sampling (LHS)

Latin Hypercube Sampling is a stratified random sampling technique that was used in this study to generate a comprehensive set input parameters, in this case the structural dimension parameters, to ensure that the sample inputs are spread across the entire distribution of the parameter space and allow the surrogate model to output, or "understand" what occurs, at each combination of design parameters. Using the generated samples, a FEA simulation can be run for structural design optimization, and the input and output variables can be recorded.

Using the newly found data points, a surrogate can be fitted and the structural optimization problem of minimizing weight but maximizing the experienced von Mises stress without exceeding the allowable stress limit can be solved by following

the trend of the sampled data points and their responses.

Surrogate Modeling – Kriging

The Kriging surrogate method is an interpolating, geostatistical model that utilizes discrete sample points of data from an experiment and interpolates the model exactly through the experimental function values under the assumption that the data used is noise free, by using the mean of the Gaussian process of the data point. The method uses a trend function or a global function and often uses lower order polynomials to fit the data, and the local departure term acts similar to an error when plotting prediction points. Equation 1 describes the Kriging formula where $\hat{y}(\boldsymbol{x})$ is the output, $\boldsymbol{\xi}(\boldsymbol{x})^T$, is the trend function, $\boldsymbol{\beta}$, are the global function parameters/coefficients, and $s(\boldsymbol{x})$ is the local departure.

$$\hat{y}(\boldsymbol{x}) = \boldsymbol{\xi}(\boldsymbol{x})^T\boldsymbol{\beta} + s(\boldsymbol{x}) \tag{1}$$

The Kriging surrogate assumes that the predictions are inversely proportional to the distance between the data points. Therefore, if two data points are closer the Kriging surrogate determines those data points are more closely correlated [3] [4].

Surrogate Modeling – Polynomial Response Surface (PRS)

The PRS is a regression-based curve fitting method where a continuous function is desired to be determined from a set of limited and noisy discrete data. The accuracy of the PRS increases with the number and location of samples. Equation 2 describes the formula of the PRS surrogate where $\hat{y}(\boldsymbol{x}, \boldsymbol{\beta})$ is the algebraic function approximation of the response surface, $y(\boldsymbol{x})$. ϵ is the approximation error between the observation and the true function.

$$y(\boldsymbol{x}) = \hat{y}(\boldsymbol{x}, \boldsymbol{\beta}) + \epsilon \tag{2}$$

The goal of PRS is to determine the global function parameters/coefficients, $\boldsymbol{\beta}$, to minimize the error when fitting the experimental data points.

Cross Validation

To assess evaluate the surrogate's ability to generalize to new data, the total dataset of sample points was split into multiple subsets for training, and then subsets to test the surrogate's ability to predict results. For this study, the cross-validation error e_{RMS} was required to be less than 5% of the range of the function.

II. Procedure

In this structural optimization study, a distributed load along the bottom length of the beam was assumed to be applied along the bottom length of the full beam, and the resulting MBB beam shown in Figure 3 was modeled in O. Sigmund's 99-line topology optimization script with the given material and dimension parameters, displayed in Table I.

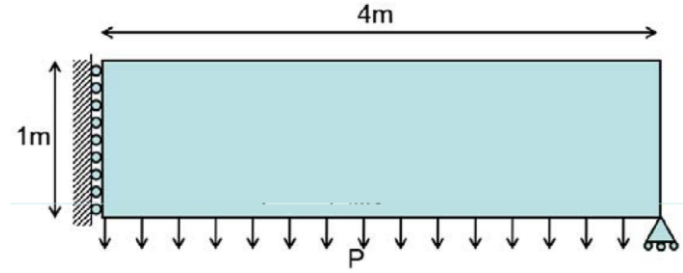

Fig. 3. The MBB beam model used for this structural optimization study.

TABLE I
INITIAL DESIGN PARAMETERS

Variable	Value	Units
Height	1	m
Length	4	m
Thickness	0.02	m
Density	7835	kg/m^3
Young's Modulus	200	GPa
Poisson's Ratio	0.3	-
Allowable Stress	250	MPa
Distributed Force	20	kN/m
Volume Fraction	0.4	

Topology Optimization Study

Three different mesh ratios were tested for the topology optimization output. Ratios of 40x10, 80x20, and 160x40 were simulated. Based on the mesh study the 40x10 ratio did not provide enough resolution to have a clear geometry that could be interpreted well. The 80x20 ratio had the simplest interpretable geometry whereas the 160x40 ratio included two more truss beams which would have increased the geometry complexity, thus increased the difficulty of manufacturing and potentially increased the complexity of the design parameters as specific lengths would have ended at different vertices instead of the main outer truss structure. Therefore, the structure produced by 80x20 was utilized for this study. Figures 4-6 show the results of the topology study.

Fig. 4. 40x10 Mesh Ratio

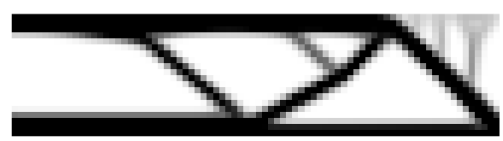

Fig. 5. 80x20 Mesh Ratio

Fig. 6. 160x40 Mesh Ratio

Ansys Workbench - Discovery

The chosen geometry from the 80x20 mesh ratio was then modeled into Ansys Workbench's geometry software, Discovery, by interpreting the pixels coordinates as a fraction of the dimensions of the beam to create the sketch. The number of pixels correlated to the thickness of the structure. A thickness of 0.02 m was added to the sketch.

Figure 7 shows the final geometry used for FEA analysis. In Discovery, certain beam widths were chosen as the dimensional design parameters, indicated by the labels in Figure 7. Using the Pull function in the software, the design parameters were created and conveniently, Discovery had an option to generate Latin Hypercube Sampling for the design parameters in which all the values were set from their initial values as the largest thickness, to the thinnest thickness that the FEA element could represent, which was 25 mm. The black markers indicate which truss beams each variable is referring to. The top truss beam was only able to be edited at the specified length due to the Ansys Discovery software not recognizing the full length of the top truss.

These parameters were chosen due to their non-conflicting locations against the support locations. Initially, the bottom truss beam was chosen to be a parameter as most of the load is taken at the lower beam location, however, due to Ansys not recognizing the geometry changes that were also occurring due to the Left Inner Truss Beam thickness changes that also inherently changed the bottom truss beam thickness due to the corners sharing the vertices, that beam was not able to be selected. Upon learning this software constraint, the Right Outer Truss Beam was chosen for its location as it did not conflict with a support location and its geometry location of not sharing the area with another design parameter. The lengths of beams were not chosen as the points of contact of the supports are also relatively at the corners of the original beam shape, which did not allow for much material reduction as those locations were a criteria not a constraint.

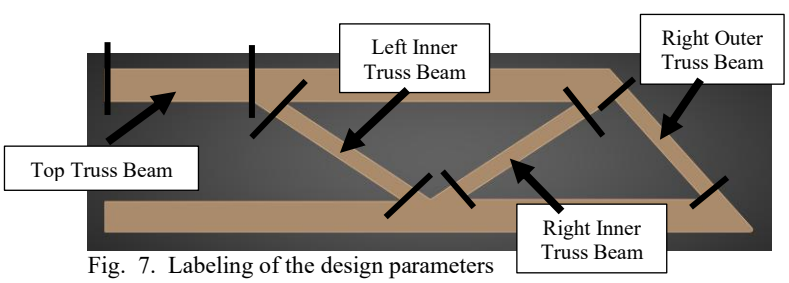

Fig. 7. Labeling of the design parameters

Ansys Workbench – Mechanical

Ansys Mechanical was used to complete the FEA analysis with the design parameters. The mesh was created with a 25 mm element size with the hex dominant method to model the structure. The element size of 25 mm was chosen as the student version of this license was only able to support 32,000 nodes/elements, as well as balancing the speed of the analysis. The 25 mm element size hit the balance between computational limitation and analysis convergence speed, and the mesh is shown in Figure 9.

Using Figure 8 to describe the placement of the supports, location 1 and 3 used compression only supports, and location 2 used a frictionless support to model the MBB beam roller supports while also staying within Ansys's constraints of maintaining 6 DoF restriction at a support. A compression only support on the front and back face of the structure was placed to ensure the deformation only occurred in the Y-axis.

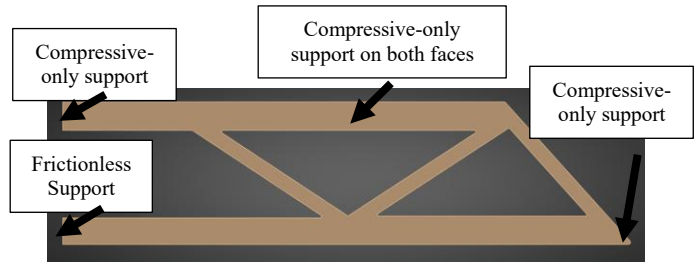

Fig. 8. Support locations

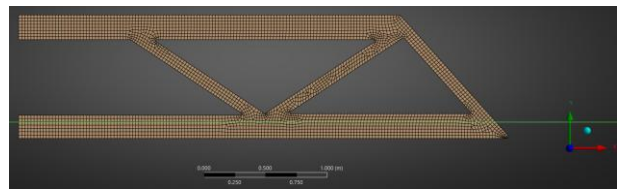

Fig. 9. Mesh used for FEA for the Latin Hypercube Sampling generated design parameters.

The line force was placed along the bottom edge of the structure. As this has thickness, the line force magnitude was split into two with each bottom edge carrying 10 kN/m, essentially totaling 20 kN/m on the bottom edge. The design parameters and the output weight and maximum von Mises stress was saved as parameters and recorded onto a .xlsx file for use in MATLAB.

MATLAB Surrogate Simulation

The Kriging model was fit using MATLAB's fitgrp() function, and the PRS model was fit using the fitlm(). The input parameters were optimized for minimal weight and maximum von Mises stress. The cross-validation method used was the Leave-One-Out Cross Validation (LOOCV) method, where K equals the number of data points, essentially making the number of training data points the same as the number of testing data points. This was chosen because the number of data points is smaller, due to the limited computation power producing less than 20 output scenarios. The design parameters found were then implemented into the FEA analysis to verify the results.

III. Results

The initial design parameters are listed in Table II with four thicknesses of the trusses indicated in Figure 7. The maximum von Mises stress experienced was 47 MPa and the mass was 281 kg at the first iteration. Figure 10 shows the first iteration of the topology optimized geometry. Figure 11 shows the first iteration of the stress analysis completed by Ansys Mechanical.

TABLE II
First Iteration Results

Parameter	Value	Units
Left Inner Truss Beam	100	mm
Right Inner Truss Beam	100	mm

TABLE II
FIRST ITERATION RESULTS

Parameter	Value	Units
Top Truss Beam	200	mm
Right Outer Truss Beam	200	mm
Maximum von Mises Stress	47	MPa
Mass	281	kg
Volume Total	0.0339	kg/m^3

Note: Volume of the structure was recorded which was then converted to mass.

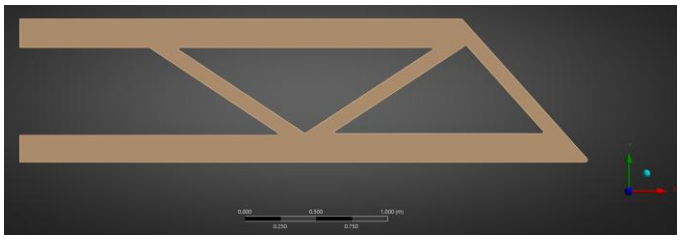

Fig. 10. First Iteration of topology optimized geometry

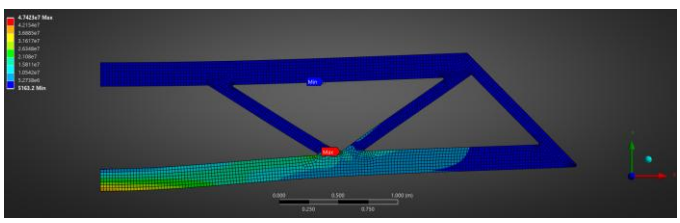

Fig. 11. First iteraton FEA stress simulation results

The final number of Design Points used in Ansys for the parametric design study was 16 points, each Design Point having 4 design parameters and two output parameters of weight and stress. 4 Design Points were corrupted during the FEA simulation with optimized design parameters.

The final design parameters from the Kriging simulation are listed in Table III. The design parameters, the predicted weight and stress, and actual weight and stress are listed. Figure 12 shows the first iteration of the topology optimized geometry. Figure 13 shows the first iteration of the stress analysis completed by Ansys Mechanical.

TABLE III
FINAL KRIGING OPTIMIZATION RESULTS

Parameter	Value	Units
Left Inner Truss Beam	27.3951	mm
Right Inner Truss Beam	62.148	mm
Top Truss Beam	30.8884	mm
Right Outer Truss Beam	44.9156	mm
Predicted Max. von Mises Stress	72.8817	MPa

TABLE III
FINAL KRIGING OPTIMIZATION RESULTS

Parameter	Value	Units
Predicted Mass	189.2	kg
Predicted Volume	0.0241518	kg/m^3
Actual Max. von Mises Stress	73.885	MPa
Actual Mass	184.9	kg
Actual Volume	0.0236	kg/m^3

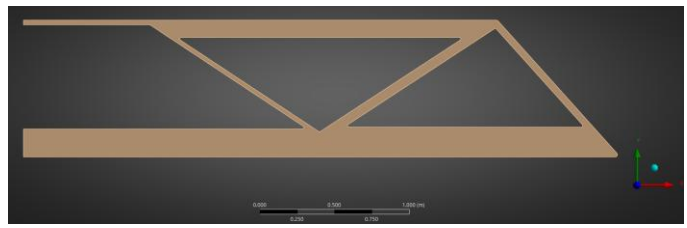

Fig. 12. Kriging Optimization Final Iteration Geometry

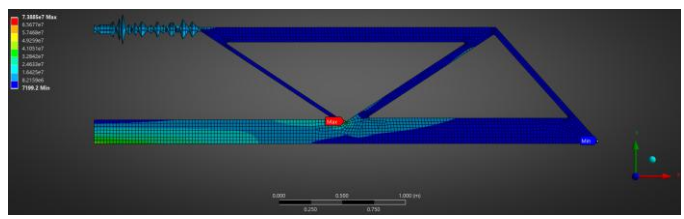

Fig. 13. Kiriging Optimized Final Iteration Geometry FEA stress simulation results

The final design parameters from the PRS simulation are listed in Table III. The design parameters, the predicted weight and stress, and actual weight and stress are listed. It should be noted that the Right Outer Truss Beam was thinner for the PRS optimization compared to the Kriging optimization, while the other dimensions were relatively close to each other. Figure 14 shows the first iteration of the topology optimized geometry. Figure 15 shows the first iteration of the stress analysis completed by Ansys Mechanical.

TABLE IV
FINAL PRS OPTIMIZATION RESULTS

Parameter	Value	Units
Left Inner Truss Beam	27.3951	mm
Right Inner Truss Beam	62.148	mm
Top Truss Beam	30.8884	mm
Right Outer Truss Beam	32.2818	mm

TABLE IV
FINAL PRS OPTIMIZATION RESULTS

Parameter	Value	Units
Predicted Max. von Mises Stress (PRS)	81.0952	MPa
Predicted Mass (PRS)	170	kg
Predicted Volume (PRS)	0.0217377	kg/m^3
Actual Max. von Mises Stress	72.314	MPa
Actual Mass	183.3	kg
Actual Volume	0.0234	kg/m^3

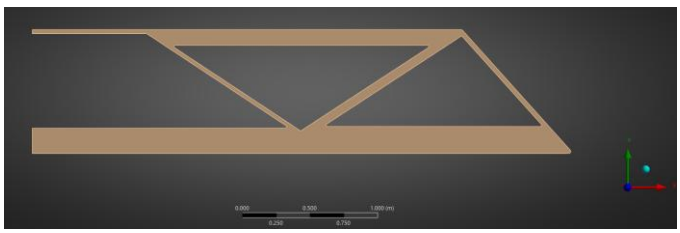

Fig. 14. PRS Optimized Final Iteration Geometry

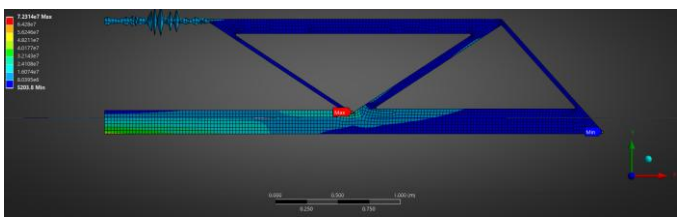

Fig. 15. PRS Optimized Final Iteration Geometry FEA stress simulation results

The objective function as a function of design iteration is displayed in Figure 16. The objective function converged within 10 iterations of the simulation for volume calculations.

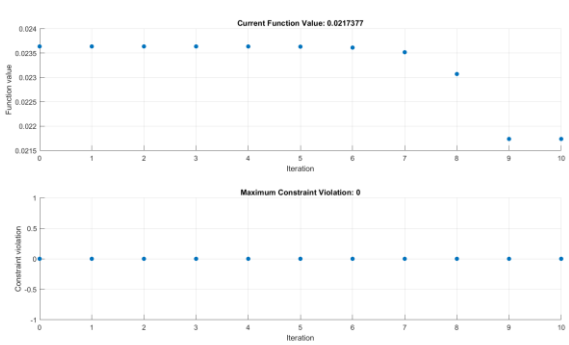

Fig. 16. Graph of objective function with Volume vs. Iteration Count

IV. DISCUSSION

Design Parameters – First Iteration vs. Surrogate

The FEA results of the first iteration geometry were found to have a maximum experienced von Mises stress of 47 MPa and a volume of 0.0339 kg/m^3 resulting in a mass of 281 kg. Using the Kriging optimization method, the design parameters of the four thickness dimensions were reduced by 72.6%, 37.9%, 84.6% and 77.5% respectively in the order of Left Inner Truss Beam, Right Inner Truss Beam, Top Truss Beam, and Right Outer Truss Beam. The volume was found to be 0.0236 kg/m^3 resulting in a mass of 184.9 kg, 34.2% lighter, and a maximum experienced von Mises stress 73.885 MPa, a 36.4% increase .

Using the PRS optimization method, the design parameters of the four thickness dimensions were reduced by 72.6%, 37.9%, 84.6% and 83.9% respectively in the order of Left Inner Truss Beam, Right Inner Truss Beam, Top Truss Beam, and Right Outer Truss Beam. The volume was found to be 0.0234 kg/m^3 resulting in a mass of 183.3 kg, 34.8% lighter, and a maximum experienced von Mises stress 72.314 MPa, a 35% increase.

Based on these observations, it could be argued that the Kriging surrogate is a more conservative model in this case due to the larger thickness prediction for optimizing for minimal material while maximizing experience von Mises stress up until the allowable stress limit.

Design Parameters – Kriging vs. PRS

Coincidentally, the first three input design parameters of the thicknesses for Left Inner Truss Beam, Right Inner Truss Beam and Top Truss Beam were the same. For the Right Outer Truss Beam thickness parameter, the Kriging model predicted 44.9156 mm whereas the PRS model predicted 32.2818 mm, a 28.1% decrease from Kriging to PRS prediction.

V. REFERENCES

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